



Every device with networking capabilities is assigned a unique MAC address.

is known as the device IP address. This IP address is in effect the name of the device on the network. The IP address is a 32-bit number which is formatted in a “dot address” format which has four sets of numbers that are separated by a period e.g. 10.240.162.178. In this format each of the numbers is representative of an eight digit binary which is a series of 0s and 1s. The first set is the network address while the remaining sets are the address of the specific host device.

The IP address is required for a network router to determine which data needs to be sent to the specific device. The protocol works to transmit the data to the router with a specific IP number or address which is the destination device’s network name. The router is then able to redirect the file to the recipient’s IP address device. In the current state of the Internet, the majority of IP numbers are those of computers and servers. But in the age of the Internet of Things, every connected device would require an IP address. Under the current system of IP address format (version four or IPv4) there are about 4.3 billion unique numbers, most of which have already been assigned. In the next generation of IP address format i.e. IPv6 the number of address increases to a potential max of 240 trillion trillion or 240 undecillion (US format), which is 240 followed by